PRODUCT MANUAL

Running & Milling Manual - ACT-FP-5500 'Liberty' Composite Frac Plug

Document MAN-FP-5500 | Revision E | 2025-12-10

Product Family	ACT-FP-5500
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Document Control	L. Hargrove
Distribution	Field operations, service centers, licensed distributors

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Product Revision History (from engineering records)

Rev	Date	Description of Change	By
A	2022-11-08	Initial release	R. Callahan
B	2023-05-16	Ceramic button count 7 to 9 per slip	A. Gutierrez
C	2024-01-24	Dissolvable Mg ball made standard	R. Callahan
D	2024-09-19	Phenolic ball option -08-P added	A. Gutierrez
E	2025-11-20	Shear screws qty 6 to 8 per ECO-2025-014; set force 55,000 lbf	R. Callahan

1. GENERAL

This manual covers wireline deployment, frac operations and mill-out of the ACT-FP-5500 'Liberty' composite frac plug for 5-1/2 in. casing (17-23 lb/ft). The plug is rated 10,000 psi differential at 300 degF and is set on the ACT-ST-10 (#10) setting tool with the ACT-FP-AK-55 adapter kit. Rev E incorporates ECO-2025-014: eight (8) 1/4-28 brass shear screws giving a 55,000 lbf setting force.

NOTE: Plugs shipped before 2025-11-20 (Rev D and earlier) use six shear screws and a 42,000 lbf set force. Verify revision from the crate label before programming the setting tool charge.

2. SAFETY

WARNING: Explosive hazard. The ACT-ST-10 setting tool uses a #10 power charge. Only licensed explosives handlers may arm the tool. Follow API RP 67 wellsite procedures.

WARNING: Do not exceed 250 ft/min wireline running speed below the last restriction; fluid bypass around the plug can pre-set the element.

- Confirm well is static or pump-down rates agreed with wireline supervisor.
- Frac ball is dissolvable Mg alloy: keep dry in sealed bag until loading.
- Composite dust from mill-out is an irritant - use cuttings management.

3. PLUG PREPARATION

Step	Action
1	Unpack plug; verify part number ACT-FP-5500 and Rev E label; inspect element for cuts.
2	Verify 8 brass shear screws installed and flush (Rev E). Do not add or remove screws.
3	Attach ACT-FP-AK-55 setting sleeve and tension rod to ACT-ST-10; torque rod to 90 ft-lb.
4	Verify shear stud engagement 0.62 in. minimum thread engagement.
5	Record plug serial number, stage number and setting depth on the run sheet.

4. PUMP-DOWN AND SETTING

Pump down at 6-12 bbl/min maximum, tracking wireline tension. Slow to 3 bbl/min for the last 500 ft. On depth, stop pumps, correlate CCL, and fire per the wireline operator's procedure. The oil-metered ST-10 stroke sets the plug in 25-40 seconds; tension drop confirms shear release from the adapter rod.

Parameter	Value
Max pump-down rate	12 bbl/min
Setting force (Rev E)	55,000 lbf nominal
Shear release after set	28,000 - 32,000 lbf
Post-set pressure test	Per frac design, typ. 1,000 psi over frac gradient

5. FRAC BALL OPERATION

Drop the 1.75 in. dissolvable Mg ball and displace at 8-10 bbl/min. Ball-on-seat is indicated by a sharp pressure increase; do not exceed 10,000 psi differential across the plug. In wells below 150 degF BHST, use the phenolic ball option (ACT-FP-5500-08-P) because the Mg ball dissolution rate is temperature-dependent (guideline: 60% mass loss in 36 hr at 200 degF in 3% KCl; 8 days at 120 degF).

6. MILL-OUT

Parameter	Recommended
Mill type	4.50 in. tri-blade junk mill, crushed carbide
RPM	80 - 120 (motor) / 40 - 60 (rotary)
WOB	1,500 - 3,500 lbf
Circulation	3+ bbl/min with sweeps each plug
Average mill time per plug	11.4 min (field average, n = 612)
Max mill time before review	25 min - stop and evaluate BHA

Composite cuttings float; ensure adequate annular velocity or reverse circulation capability. Slip button fragments (ceramic) report to surface as sand-sized grit.

7. QUALITY RECORDS

Each production lot is qualified per API 19V Grade V3. Incoming composite mandrel stock is inspected per QIR series (see QIR-2024-0871 for an example lot record). Setting force verification is performed on one plug per 200 manufactured.

Document Approval

Role	Name	Signature	Date
Prepared	A. Gutierrez	*/signed electronically/*	2025-12-10
Quality Review	S. Nguyen	*/signed electronically/*	2025-12-10
Approved	J. Martinez, P.E.	*/signed electronically/*	2025-12-10